

# Bulk RNA-seq Analysis of Type 2 Diabetic vs Control Skin

## Presentation

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# Introduction & Workflow Overview

## Introduction

- Type 2 diabetes affects multiple tissues, including the skin.
- Using RNA-seq, we are able to measure how gene expression changes in disease
- The study used for the purpose of this experiment compares skin tissue from Type 2 diabetic patients and non-diabetic controls.

## Workflow Overview

- Data Download (SRA dataset)
  - Quality Control (FastQC & MultiQC)
  - Read alignment (Star alignment to human genome)
  - Quantification (featureCounts)
  - Differential Expression (DESeq2)
  - Functional Enrichment (GO/KEGG)
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# Data and Experimental Setup

Article: Genome wide analysis of gene expression changes in skin from patients with type 2 diabetes

## Dataset:

- GSE144441 from NCBI GEO

## Organism:

- Homo sapiens (Human)

## Samples:

- 15 diabetic skin tissues vs 12 control skin tissue

## Goal:

- Identify DEGs and enriched Pathways  
Perform functional enrichment to find affected pathways

## Why This Dataset:

- It was a well annotated dataset
- It has balanced group samples
- Relevant to understanding diabetic skin complication



# Quality Control



## Raw Data Inspection:

- Used FastQC to evaluate raw FASTQ files
- Checked per-base sequence quality, GC content, adapter contamination, etc.

## Combined results using MultiQC

- Allowed visualization of QC metrics across all 27 samples

## Results:

- High sequencing quality ( $>Q30$ )(Figure 1)
- High duplication (normal for RNA-seq)
- Mild adapter contamination, but trimming wasn't required
- All samples passed QC for downstream analysis

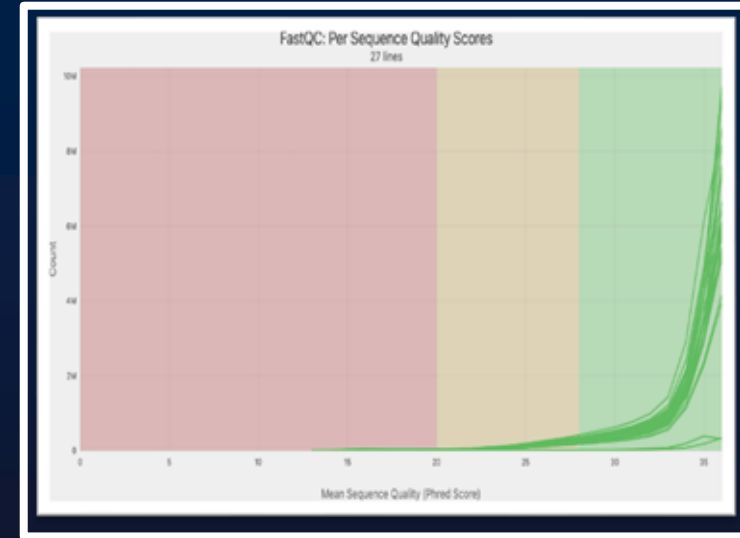
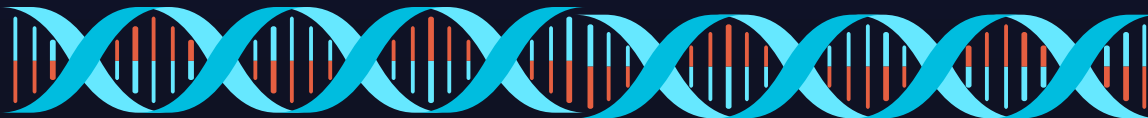


Figure 1: Per sequence quality score plot

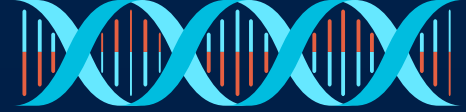




# Differential Gene Expression Analysis (DEG)



# MA Plot



- The MA plot (**Figure 2**) visualizes the relationship between gene expression strength (mean normalized counts) and log2 fold change between diabetic and control samples.
- Most genes cluster around  $\log_2FC = 0$ , meaning they are not differentially expressed.
- The blue points highlight genes that are significantly up- or down-regulated in Type 2 diabetic samples.
- We observe more significant downregulated genes than upregulated ones,
- Which tells us that there is a reduced expression of several genes in diabetic tissue.

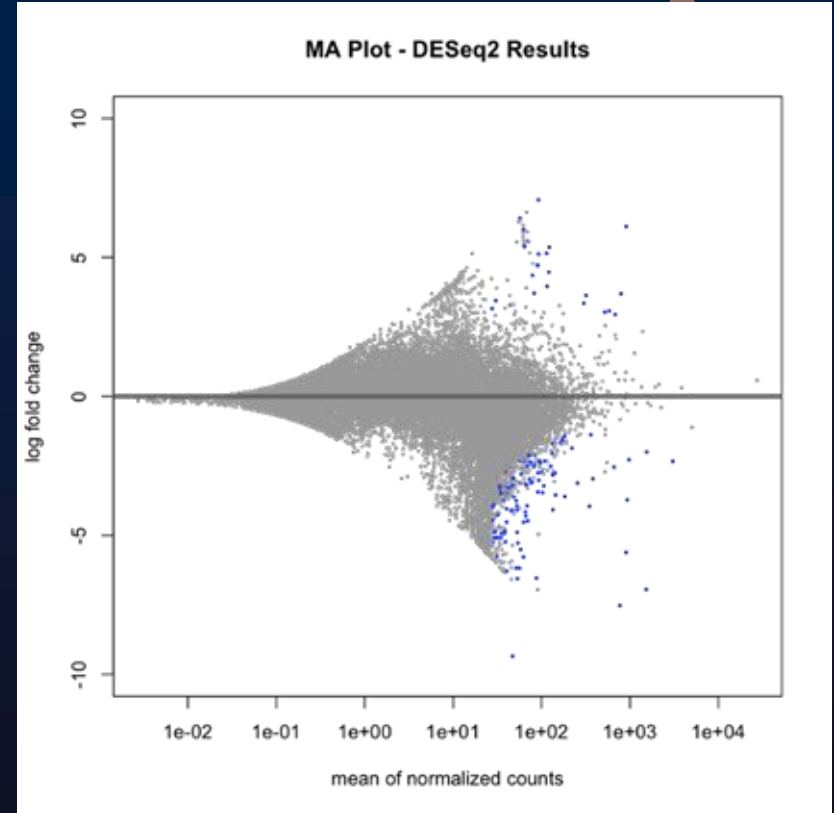
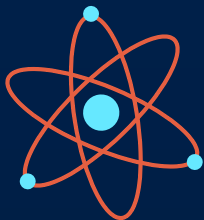


Figure 2: MA Plot



# Volcano Plot

- This volcano plot (**Figure 3**) highlights which genes change the most and which changes are statistically meaningful
- Purple points represent genes that meet the significance threshold (adjusted  $p < 0.05$  and  $|\log_2\text{FC}| > 1$ ).
- Which tells us that several genes shows strong signals, distinguishing diabetic from control samples

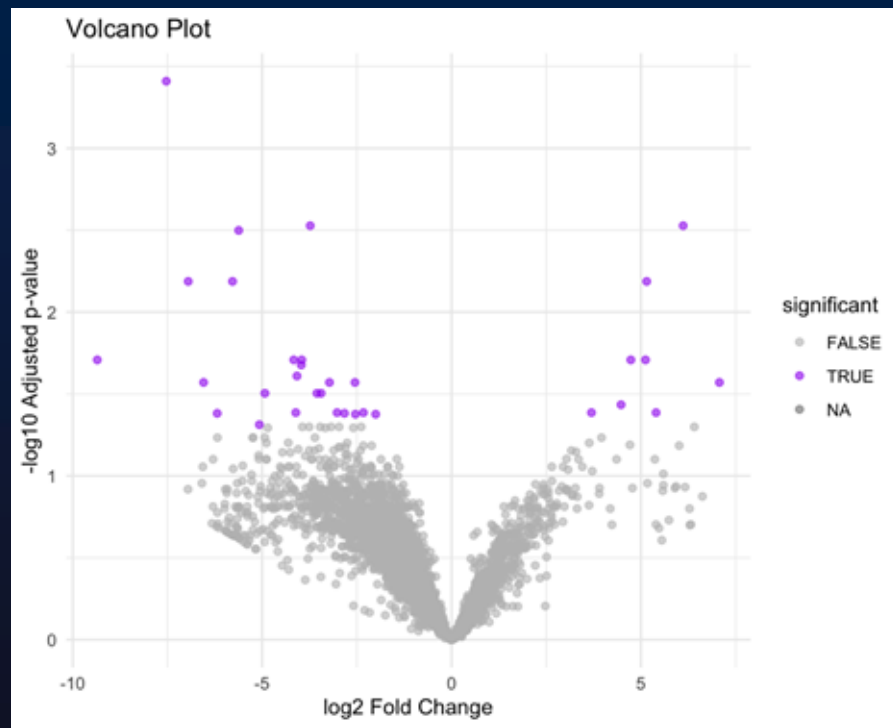


Figure 3: Volcano Plot







# Key DEGs

## How Key DEGs were identified

We used DESeq2 to perform differential expression analysis

The applied thresholds are:

- Adjusted P-value  $< 0.05$
- $|\text{Log}_2 \text{ Fold Change}| > 1$

Extracted top upregulated and downregulated genes for closer inspection

## Most Strongly Down Regulated Gene

ENSG00000204427 (ABHD16A)- Largest negative log2 fold change

Genes involved cell adhesion, skin structure and ECM organization

## Biological Relevance

Downregulated junction/adhesion genes = weaker cell-cell connections

Fits clinical diabetic symptom: Slow wound healing & high infection risk

Large expression shifts indicate major structural changes in diabetic skin



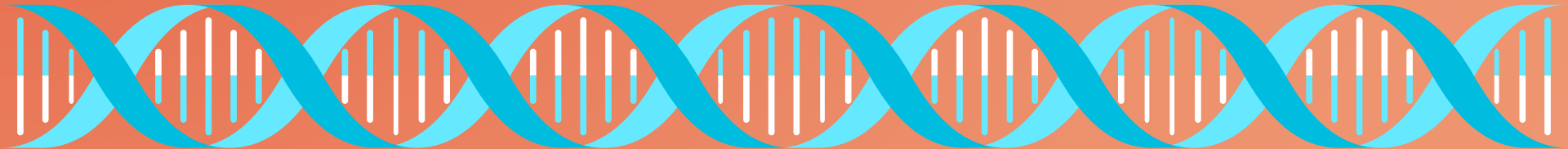
# Major enrichment pathways



**Adherens junction pathway** was significantly enriched among the differentially expressed genes (DEGs).

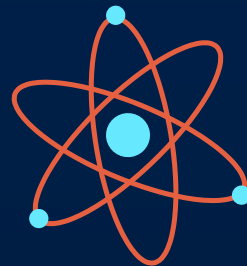


Changes in genes responsible for maintaining **cell-cell adhesion** highlight a potential mechanism underlying **skin fragility and dysfunction in diabetes**.





# Biological Interpretation



## Altered Cell-Cell Adhesion & Tissue Integrity

- Down Regulation of adhesion and cytoskeleton related genes (e.g. NF2, SYNPO)
- This may contribute to reduced skin barrier



## Disturbed Metabolic Regulation

- Changes in metabolic genes (PFKP, PLIN4, ABHD16A) indicate disturbance in glucose and lipid metabolism



## Impaired Wound Healing and Tissue Structure

- The combined effects on adhesion pathways, cytoskeletal organization, and inflammation related genes (INHBA, HSF2) are consistent with poor tissue remodeling and slower wound healing

# Conclusion



## Downregulation

Diabetic skin shows major downregulation of expressed genes



## Adhesion

Structural and adhesion pathways are disrupted



## Biological

Impaired wound healing and increased infection risk

All these provide a molecular explanation for skin complications in Type 2 diabetes



**Thank you  
Questions?**